

Job Finder (Placement Portal Application) – MAD-1 Project

1. Student Details

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I am pursuing BS in Data Science from IIT Madras and currently in diploma level.

2. Project Details

Project Title: JobFinder

This project belongs to Modern Application Development -I course, which is the part of IITM BS data science program. The Job Finder (Placement Portal Application) is a web-based system which automates campus recruitment within the institute. It will manage the interaction between Admin, Companies, and student through role-based access. This project will allow:

Admin is super user who can manage the company and students.

Companies can register themselves, post the jobs, and track the applied student applications and update their status.

Student can see all registered companies and their jobs posting and apply for different jobs based on eligibility criteria, and check the status of applications and history of applied jobs.

3. AI/LLM Declaration

I used ChatGPT to get suggestions in model, backend logic, some HTML, CSS layout, and improving variable naming consistency.

The extent of AI/LLM usage is around 10-12%, limited to code suggestions and some logic building.

All the final implementation logic, debugging and integration were done manually.

4. Technology Used

- **Flask:** - It is used to build the server-side logic of the application.
- **Flask_Sqlalchemy:** - It provides a straightforward way to work with databases using Python objects.
- **Jinja2:** - It is used to render dynamic HTML pages from the flask backend.
- **SQLite:** - It is used as the database to store the data.
- **HTML:** - It provides the structure of pages.
- **CSS:** - It is used for styling and improving the visual appearance.
- **Bootstrap:** - It provides quick building responsive UI.

5. DB Schema Design

There are total 5 tables. It includes features like role-based access of Admin, Company, and Student and timestamps for tracking records. Foreign key and Enum ensure data integrity and role management. All the relationships are properly defined with foreign keys.

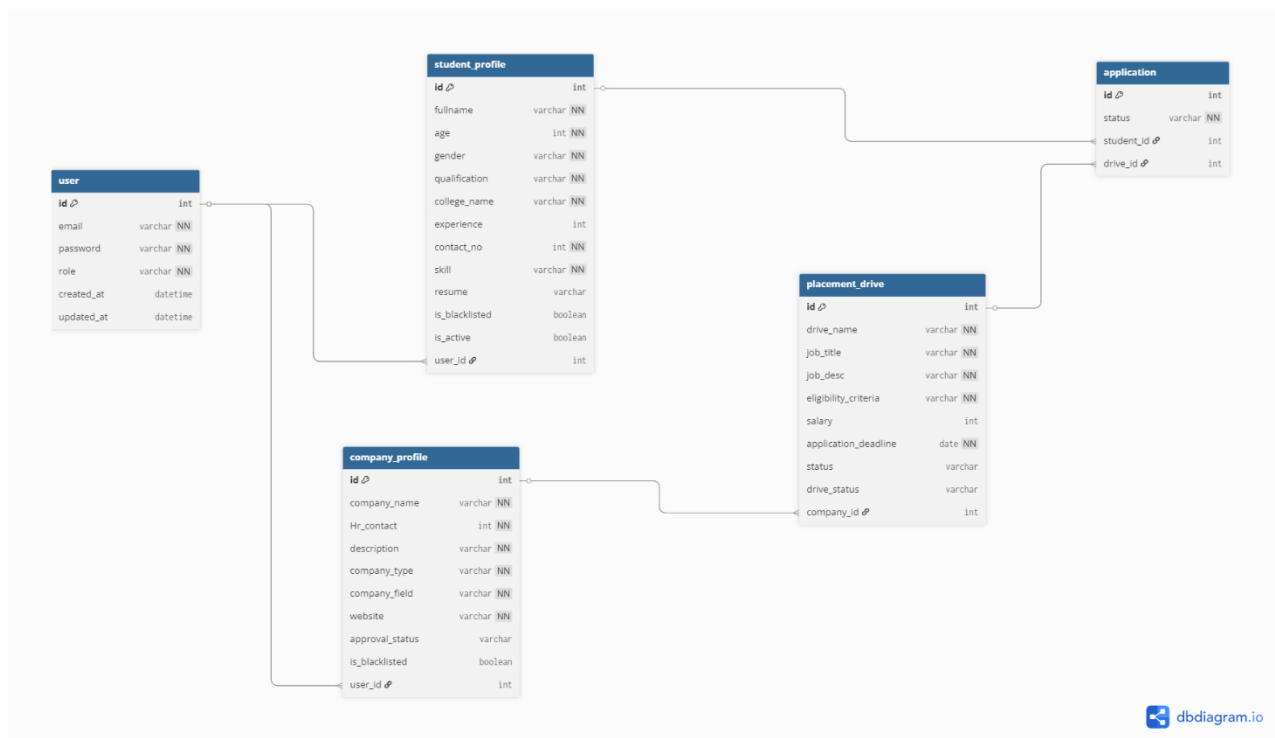
Table used in Database:

- User
- Student_Profile
- Company_Profile,
- Placement_Drive
- Application

Relationship:

- **One -to-One:** User -> Student_Profile, User -> Company_Profile,
- **One-to-Many:** Student_Profile -> Application, Company_Profile -> Placement_Drive, Placement -> Application
- **Many-to-Many:** Student_Profile <-> Placement_Drive (through Application)

ER Diagram



6. Architecture and Features

app.py - It contains flask code to initialize the app.

/controllers.py - it contains all flask routers.

/models.py - It contains all database models using SQLAlchemy.

/style.css - It is in static folder in which contains all CSS files.

/resumes - It is in static folder in which contains all .pdf format files which is uploaded by student.

/templates - It contains all HTML, Bootstrap codes for Admin, Company, Student, Login and Signup

Implemented Features:

- User login (Admin, Company, Student) and Registration (Company, Student) and cannot register with same email id if register it before.
- Secure password hashing.
- User session handling and role-based access.
- Student can not apply again in same drive if applied before.
- Admin is superuser and see the total number of register company, placement drive, student, applications, and approve the company and drive, block and unblock the student and company, see the ongoing drives and student application and their resume.
- Student can apply in many drives, check the status, Upload resume.
- Company can create drives, Edit, Delete and Mark as complete after approval by admin, track and update the student application.

7. Video and GitHub

Demo video: -

[https://drive.google.com/file/d/1Vkj7gomHHWlf4wHIM8GqBMclRWloh70U/view?usp=drive link](https://drive.google.com/file/d/1Vkj7gomHHWlf4wHIM8GqBMclRWloh70U/view?usp=drive_link)

Project GitHub: -

<https://github.com/22f3001353/jobfinder>